

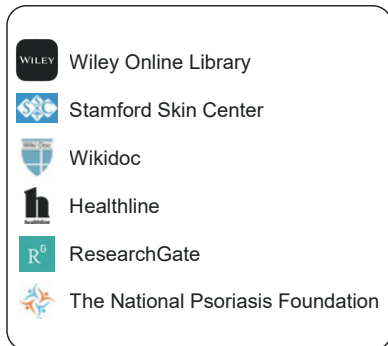
a

Construction of Rare Skin Disease Dataset (RSDD)

Data Sources

Disease Selection Criteria

Categories and Sample Size of Final Dataset (n=564)



Rare Disease Diagnosis and Treatment Guidelines



Blue Rubber Bleb Nevus Syndrome (73)



Epithelioid Sarcoma (46)



Behcet's Disease (81)



Generalized Pustular Psoriasis (110)



Cutaneous Neuroendocrine Carcinoma (115)



Cryopyrin-associated periodic syndrome (22)



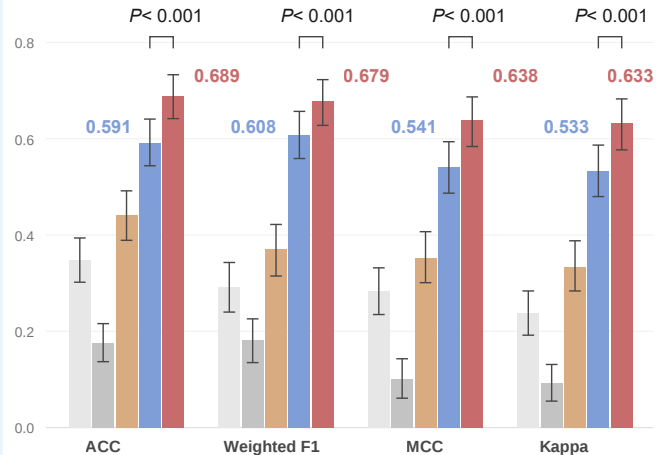
Dermatofibrosarcoma Protuberans (59)



Gorlin Syndrome (58)

b

Performance Comparison on RSDD



MedGemma Hulu-Med Qwen3-VL PanDerm SkinGPT-X

c



Primary Diagnosis: Blue Rubber Bleb Nevus Syndrome

Key Findings: The image shows multiple discrete, well-defined, dark brown to black macules and papules on the upper chest and neck, set against a background of diffuse erythema. The lesions are smooth, lack scaling, crusting, ulceration, or telangiectasia, and have regular borders. The surrounding erythema suggests a separate inflammatory process. The lesions appear stable and non-infiltrative.

Case Reviewing: While the description doesn't explicitly mention a 'blue' hue, the dark pigmentation and morphology are consistent with melanocytic lesions. The presence of multiple lesions, particularly the linear arrangement of three lesions below the clavicles, raises suspicion for a syndrome. The surrounding erythema is noted as a separate process. **Comparing these findings with the provided diagnostic guidelines - Blue Rubber Bleb Nevus Syndrome:** The description aligns well with the morphology of blue rubber bleb nevi (dark pigmented papules/macules, smooth surface, well-defined borders). Although the classic color is 'blue', dark brown/black pigmentation can occur, and the syndrome involves multiple lesions.



Primary Diagnosis: Behcet's Disease

Key Findings: The lesion depicted is a solitary, well-demarcated, erythematous ulcerative lesion located on the buccal mucosa. Morphologically, it presents as a shallow, irregularly bordered ulcer with a central area of necrotic debris and surrounding erythema.

Case Reviewing: The visual description in detail an erythematous, ulcerative lesion on the buccal mucosa with irregular borders, central necrosis, and surrounding inflammation. This morphology strongly aligns with the typical presentation of oral ulcers in Behcet's disease. **The diagnostic guidelines for Behcet Disease mention ulcers, asymmetry, lack of internal patterns, and inflammation, all consistent with the findings. Past cases also describe similar oral lesions consistent with Behcet's Disease.** The other options are less likely: Gorlin Syndrome typically involves basal cell carcinomas, not oral ulcers; Blue Rubber Bleb Nevus Syndrome involves specific blue papules and systemic features not described. Therefore, Behcet's disease is the most fitting diagnosis based on the visual findings, model prediction, and comparison with standard references and past cases.